

Flatten#

[https://pytorch.org/docs/stable/generated/torch.nn.modules.flatten.Flatten.h](https://pytorch.org/docs/stable/generated/torch.nn.modules.flatten.Flatten.html)

Description:

Flattens a contiguous range of dims into a tensor. For use with Sequential, see torch.flatten() for details. Input: $(*, S_{\text{start}}, \dots, S_i, \dots, S_{\text{end}}, *)$, where S_i is the size at dimension i and $*$ means any number of dimensions including none. Output: $(*, \prod_{i=\text{start}}^{\text{end}} S_i, *)$. start_dim (int) – first dim to flatten (default = 1).

Function Signature

```
class torch.nn.modules.flatten.Flatten(start_dim=1,
end_dim=-1)[source]#
```

Shape:

Parameters

Examples::

```
extra_repr()[source]#
```

Returns:

Tensor

Code Examples

```
>>> input = torch.randn(32, 1, 5, 5)
>>> # With default parameters
>>> m = nn.Flatten()
>>> output = m(input)
>>> output.size()
torch.Size([32, 25])
>>> # With non-default parameters
>>> m = nn.Flatten(0, 2)
>>> output = m(input)
>>> output.size()
torch.Size([160, 5])
```

Detailed Documentation

Documentation

Flattens a contiguous range of dims into a tensor.

For use with Sequential, see `torch.flatten()` for details.

Input: $(*, S_{\text{start}}, \dots, S_i, \dots, S_{\text{end}}, *)$ ($*, S_{\{\text{start}\}}, \dots, S_{\{i\}}, \dots, S_{\{\text{end}\}}, *$) ($*, S_{\text{start}}, \dots, S_i, \dots, S_{\text{end}}, *$), where S_i is the size at dimension i and $*$ means any number of dimensions including none.

Output: $(*, \prod_{i=\text{start}}^{\text{end}} S_i, *)$ ($*, \prod_{i=\text{start}}^{\text{end}} S_{\{i\}}, *$) ($*, \prod_{i=\text{start}}^{\text{end}} S_i, *$).

`start_dim` (int) – first dim to flatten (default = 1).

`end_dim` (int) – last dim to flatten (default = -1).

Returns the extra representation of the module.

Runs the forward pass.